

Chapter 11 Homework Answer Sheet

Name: _____ Student ID: _____ Instructor: _____ Score: _____

Part A multiple choice (18 points)

1-5 CBCCE

6-10 BAACD

11-15 EDDDD

16-18 AAC

Part B short question (20 points)

19 (8 points)

(a) (1 point) $\text{Cl}_2(\text{g}) + \text{CHCl}_3(\text{g}) \rightarrow \text{HCl}(\text{g}) + \text{CCl}_4(\text{g})$

(b) (2 points) $\text{Cl}(\text{g})$, $\text{CCl}_3(\text{g})$

(c) (3 points) step 1: unimolecular; step 2: bimolecular; step 3: bimolecular

(d) (1 point) Step 2

(e) (1 point) $\text{Rate} = k [\text{Cl}_2]^{1/2} [\text{CHCl}_3]$ OR $k_2(k_1/k_{-1})^{1/2} [\text{Cl}_2]^{1/2} [\text{CHCl}_3]$

20 (4 points)

(a) (2 points)

This is the second order reaction.

$$1/[\text{A}]_t = 1/[\text{A}]_0 + kt$$

(b) (1 point) $k = (1/[\text{A}]_t - 1/[\text{A}]_0) / \Delta t = 4.2 \times 10^{-2} \text{ M}^{-1} \cdot \text{min}^{-1}$

(c) (1 point) $t_{1/2} = 37 \text{ min}$

21 (3 points)

(a) (1 point) $[\text{A}]_t = [\text{A}]_0 e^{-kt} = 0.0250 \text{ mol} \div 2.00 \text{ L} \times e^{-(6.82 \div 1000 \times 5 \times 60)} = 0.0016 \text{ M}$ or $1.6 \times 10^{-3} \text{ M}$
after 5.0 min moles of $\text{N}_2\text{O}_5 = 2.00 \text{ L} \times 1.6 \times 10^{-3} \text{ mol/L} = 3.2 \times 10^{-3} \text{ mol}$

(b) (1 point)

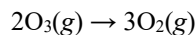
When $[\text{A}]_t = 1/2 [\text{A}]_0$, $t_{1/2} = -1/k \ln([\text{A}]_t / [\text{A}]_0) = \ln(2)/k = 0.693 \div (6.82 \times 10^{-3} \text{ s}^{-1}) = 102 \text{ s}$

(c) (1 point)

$$t = 1/k \ln([\text{A}]_t / [\text{A}]_0) = 1 \div 6.82 \times 10^{-3} \text{ s}^{-1} \times \ln(0.0250/0.0100) = 134 \text{ s}$$

22 (5 points)

(a) (1 point)



(b) (1 point)

$$\text{Rate} = k[\text{O}_3]^2 / [\text{O}_2]$$

(c) (1 point)

Intermediate

(d) (2 points)

Yes, it will change.

$$\text{Rate} = k[\text{O}_3]^2$$